ON A CONJECTURE OF DOTSENKO ON WEAK NILPOTENCE AND THE ENGEL IDENTITY

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ABSTRACT. By means of AI-assisted computer algebra, we give examples of commutative but nonassociative algebras satisfying the identity $t_4 = 0$ such that neither the 7 nor 8-Engel identity holds. This resolves one conjecture of Dotsenko in the negative, and confirms a second. Conversely, we show that $t_4 = 0$ implies the 9-Engel identity. Together these show that 9 is the least integer m for which $t_4 = 0$ implies the m-Engel identity.

1. INTRODUCTION

Let A be a vector space over a field of characteristic zero, equipped with a commutative but not necessarily associative bilinear operation. One defines an n-linear operation $t_n(a_1, \dots, a_n)$ as the sum of all distinct multilinear terms of degree n that one can define using said bilinear operation. For instance:

$$\begin{aligned} t_4(a,b,c,d) = {} & a[b[cd]] + a[[bc]d] + a[[bd]c] + [ab][cd] + [ac][bd] + [ad][bc] + [a[bd]]c \\ & + [a[bc]]d + [[ab]c]d + [[ac]b]d + [[ab]d]c + [[ad]b]c + [a[cd]]b + [[ac]d]b + [[ad]c]b. \end{aligned}$$

If t_i is identically 0 for all $i \geq n$, we say that A is *weakly nilpotent of index n*. Similarly, we define the *k-Engel* identity to be the following:

$$e_k(x,y) = \underbrace{[\cdots[x\,y]\cdots y]}_{k \text{ copies of } y} = 0.$$

For $y \in A$, let $R_y \colon A \to A$ be the multiplication operator $R_y(x) = xy$. Then $e_k(x,y) = R_y^k(x)$, so A is k-Engel precisely when $R_y^k = 0$ for every $y \in A$. Equivalently, every R_y is nilpotent of index at most k. Ternary variations of these conditions appear in Yagzhev's reformulation of the Jacobian conjecture, see [1, 6] for more details[1].

Dotsenko [2] considers the case where A is allowed to be non-commutative; using computer algebra he showed that the non-commutative analogue[2] of the $t_4 = 0$ identity, $w_4 = 0$ implies the 7-Engel identity. Based on this intuition, he conjectures the following:

Conjecture 1.1. [2, Conjecture 5] Let A satisfy the $t_4 = 0$ identity. Then A also satisfies the 7-Engel identity ie., for all $x, y \in A$ one has $e_7(x,y) = 0$.

This conjecture is false. We phrase this as a theorem.

Theorem 1.2.

(1) *There exists a 17-dimensional algebra A_{17} satisfying the $t_4 = 0$ identity for which there exists $a, b \in A_{17}$ with $e_7(a,b) \neq 0$.*
(2) *There exists a 22-dimensional algebra A_{22} satisfying the $t_4 = 0$ identity for which there exists $a, b \in A_{22}$ with $e_8(a,b) \neq 0$.*

Using the algebra A_{17} above, we confirm the following conjecture of Dotsenko.

Corollary 1.3. [2, Conjecture 4.1] *Suppose A satisfies $t_4 = 0$. It is not necessarily the case that $t_5 = 0$.*

[1] This article was written some weeks before Alpöge announced a counterexample to the Jacobian conjecture in July 2026.
[2] Note that in the non-commutative world there are more distinct multilinear terms of degree 4.

The strategy behind the construction of both examples in Theorem 1.2 was to generate the free commutative, nonassociative algebra on two generators a, b. We then quotient this by the T-ideal generated by t_4. We found that $e_i(a,b)$ was nonzero in this algebra for $i = 7, 8$ respectively. We then found some smaller examples as quotients of this. The implementations were written using Claude Code.

The same strategy can be used to produce the proof of the following which can be also verified by a relatively small program in Macaulay2.

Theorem 1.4.

(1) Let A be an algebra satisfying the $t_3 = 0$ identity. Then A also satisfies the identity $e_3 = 0$.
(2) Let A be an algebra satisfying the $t_4 = 0$ identity. Then A also satisfies the identity $e_9 = 0$.

We remark that part (1) has previously been shown in [4]. The proof of both statements involves writing e_k as a sum. In the first case, this involves four terms and can be verified by hand while in the latter it involves 482 terms and requires computer algebra software to verify. In light of Theorem 1.2 the above theorem is sharp. This suggests the following modified conjecture.

Conjecture 1.5. Let $k \geq 2$ and suppose that A satisfies the $t_k = 0$ identity. Then A also satisfies the 3^{k-2}-Engel identity ie., for all $x, y \in A$ one has $e_{3^{k-2}}(x, y) = 0$.

We were unable to verify the $t_5 = 0$ version of this conjecture using the same method in a reasonable amount of time or memory, as the number of binary trees providing generators of the free algebra on two generators grows exponentially and one would need to look at trees with 27 internal binary vertices. We were able to prove partial results in that direction: we found a 111 dimensional algebra A_{111} satisfying $t_5 = 0$ but not the 13-Engel identity.

Finally, we make a comment on the methods of this paper. We believe the results illustrate that large language models can be useful coding assistants for practitioners in nonassociative algebra (and algebraic topology), enabling fast design of custom software for explicit computation of examples and counterexamples. We believe that this suits a particular genre of search-based problem, where auxiliary software used in the problem is permitted to be of arbitrary complexity and is not required to be correct, but the output can be checked by programs simple enough to be human-audited. For instance, in this case, the examples and derivation can be checked in the form of relatively short Macaulay2 files.

AI use disclosure: Claude Code (using the Opus 4.8 model) was used to produce all of the software used in the production of this document. It was not used in the writing of the paper itself other than for spell-checking purposes and the formatting of the expression in Appendix A.

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2. The main results

In this section, we prove both theorems from the introduction.

Theorem 2.1. *There exists a 17-dimensional algebra A_{17} satisfying the $t_4 = 0$ identity for which there exists $a, b \in A_{17}$ with $e_7(a,b) \neq 0$.*

Proof. The following is the multiplication table for the counterexample. Let A_{17} be the commutative nonassociative $\mathbb{Q}$-algebra with basis $b_0, \ldots, b_{16}$ and commutative product $b_i b_j = b_j b_i$ given on the

nonzero products by

$$
\begin{array}{lcl}
b_0b_0 &=& b_1 \\
b_0b_1 &=& b_2 \\
b_0b_2 &=& -\frac{1}{4}\,b_3 \\
b_0b_3 &=& -4\,b_4 \\
b_0b_5 &=& b_6 \\
b_0b_6 &=& b_7 \\
b_0b_7 &=& b_8 \\
b_0b_8 &=& -\frac{1}{2}\,b_{10} - \frac{1}{2}\,b_{11} \\
b_0b_9 &=& b_{10} \\
b_0b_{10} &=& b_{12} \\
b_0b_{11} &=& -2\,b_{12} - \frac{1}{2}\,b_{13} \\
b_0b_{12} &=& 2\,b_{14} + \frac{1}{2}\,b_{15} \\
b_0b_{13} &=& -8\,b_{14} - 2\,b_{15}
\end{array}
\qquad
\begin{array}{lcl}
b_0b_{14} &=& -\frac{1}{4}\,b_{16} \\
b_0b_{15} &=& \frac{1}{2}\,b_{16} \\
b_1b_1 &=& b_3 \\
b_1b_2 &=& b_4 \\
b_1b_6 &=& b_9 \\
b_1b_7 &=& b_{11} \\
b_1b_8 &=& b_{12} \\
b_1b_9 &=& b_{13} \\
b_1b_{10} &=& 4\,b_{14} \\
b_1b_{11} &=& -2\,b_{14} \\
b_1b_{12} &=& \frac{1}{2}\,b_{16} \\
b_1b_{13} &=& -2\,b_{16}
\end{array}
\qquad
\begin{array}{lcl}
b_2b_5 &=& -2\,b_8 - b_9 \\
b_2b_8 &=& b_{14} \\
b_2b_9 &=& b_{15} \\
b_2b_{10} &=& b_{16} \\
b_2b_{11} &=& -\frac{1}{2}\,b_{16} \\
b_3b_5 &=& -4\,b_{10} - 8\,b_{11} \\
b_3b_6 &=& -2\,b_{13} \\
b_3b_7 &=& -2\,b_{15} \\
b_3b_8 &=& b_{16} \\
b_4b_5 &=& 4\,b_{12} + b_{13} \\
b_4b_6 &=& 2\,b_{14} \\
b_4b_7 &=& \frac{1}{2}\,b_{16}
\end{array}
$$

all other products of basis vectors being 0 and extended bilinearly. The identity $t_4 = 0$ has been verified via a Macaulay2 script by showing it on basis elements [3]. Since it holds on basis elements, it holds on the whole algebra by multilinearity. The script can be found as `verify_e7.m2` in the accompanying GitHub repository [3].

Set $x = b_5$ and $y = b_0$. Then

$$e_7 = \underbrace{[\cdots[x\,y]\cdots y]}_{7 \text{ copies of } y} = \tfrac{1}{8}\,b_{16}$$

In particular the 7-Engel value $e_7(x,y) = \frac{1}{8}\,b_{16} \neq 0$, while $t_4(a,b,c,d) = 0$ for all $a,b,c,d \in A_{17}$. Thus this is a counterexample. □

Remark 2.2. The algebra A_{17} was generated as follows. First one constructs (a finite piece of) the free commutative, non-associative algebra on the elements b_0 and b_5. We quotient this algebra by the T-ideal generated by t_4. One obtains a 45-dimensional counterexample by taking a further truncation in the obvious way designed to keep e_7 identity nonzero: we kill all basis elements of degree 9 or greater or containing two or more copies of b_5. To reduce this to 17 dimensions, we test each basis element in order, construct the ideal generated by it and then quotient by that and observe whether or not the image of $e_7(b_5, b_0)$ remains nonzero in the quotient. If it does we continue the procedure. By doing this successively, we obtain the example above. This is obviously dependent on an order on basis elements and the program tested 42 different orders before arriving at the above example. This was the smallest we found, but it may not be minimal as our algorithm is certainly not exhaustive. A similar algorithm is used to produce algebras A_{22} and A_{111}.

Theorem 2.3. *There exists a 22-dimensional algebra A_{22} satisfying the $t_4 = 0$ identity for which there exists $a, b \in A_{22}$ with $e_8(a,b) \neq 0$.*

Proof. The $t_4 = 0$ and $e_8 \neq 0$ counterexample is produced similarly as in the previous theorem:

$$\begin{aligned}
b_0b_0 &= b_1 & b_0b_{19} &= -\tfrac{1}{2}b_{20} & b_1b_{19} &= \tfrac{1}{2}b_{21}\\
b_0b_1 &= b_2 & b_0b_{20} &= \tfrac{1}{2}b_{21} & b_2b_5 &= -b_9 - 2b_{10} - b_{11}\\
b_0b_2 &= -\tfrac{1}{4}b_3 & b_1b_1 &= b_3 & b_2b_9 &= b_{17}\\
b_0b_3 &= -4b_4 & b_1b_2 &= b_4 & b_2b_{10} &= b_{18}\\
b_0b_5 &= b_6 & b_1b_5 &= b_7 & b_2b_{11} &= b_{19}\\
b_0b_6 &= b_8 & b_1b_6 &= b_{11} & b_2b_{14} &= -\tfrac{1}{2}b_{20}\\
b_0b_7 &= b_9 & b_1b_7 &= b_{13} & b_2b_{16} &= \tfrac{1}{4}b_{21}\\
b_0b_8 &= b_{10} & b_1b_8 &= b_{14} & b_3b_5 &= -4b_{12} - 4b_{13} - 8b_{14}\\
b_0b_{10} &= -\tfrac{1}{2}b_{12} - \tfrac{1}{2}b_{14} & b_1b_9 &= b_{15} & b_3b_7 &= 12b_{17} + 64b_{18} + 16b_{19}\\
b_0b_{11} &= b_{12} & b_1b_{10} &= b_{16} & b_3b_8 &= -6b_{17} - 32b_{18} - 10b_{19}\\
b_0b_{12} &= b_{16} & b_1b_{12} &= -b_{17} - 4b_{18} - 2b_{19} & b_3b_{10} &= b_{20}\\
b_0b_{13} &= -b_{15} & b_1b_{13} &= -4b_{17} - 16b_{18} - 4b_{19} & b_3b_{12} &= b_{21}\\
b_0b_{14} &= -2b_{16} & b_1b_{14} &= 3b_{17} + 14b_{18} + 4b_{19} & b_4b_5 &= b_{15} + 4b_{16}\\
b_0b_{15} &= -b_{17} & b_1b_{16} &= -\tfrac{1}{2}b_{20} & b_4b_6 &= -b_{17} - 6b_{18} - 2b_{19}\\
b_0b_{16} &= \tfrac{1}{2}b_{17} + 2b_{18} + \tfrac{1}{2}b_{19} & b_1b_{17} &= b_{21} & b_4b_8 &= \tfrac{1}{2}b_{20}\\
b_0b_{17} &= -b_{20} & b_1b_{18} &= -\tfrac{1}{4}b_{21} & b_4b_{10} &= -\tfrac{1}{4}b_{21}\\
b_0b_{18} &= \tfrac{1}{4}b_{20}
\end{aligned}$$

all other products of basis vectors being 0 and extended bilinearly. As above, the identity $t_4 = 0$ has been verified by a Macaulay2 script on basis elements, and holds on all of A_{22} by multilinearity. The script can be found as `verify_e8.m2` in the accompanying GitHub repository [3].

Set $x = b_5$ and $y = b_0$. Then

$$e_8 = \underbrace{[\cdots[x\,y]\cdots y]}_{8 \text{ copies of } y} = -\tfrac{1}{16}b_{21}$$

In particular $e_8(x, y) = -\frac{1}{16}b_{21} \neq 0$, while $t_4(a, b, c, d) = 0$ for all $a, b, c, d \in A_{22}$ and so we can conclude. □

Remark 2.4. The algebra A_{22} alone is enough to refute [2, Conjecture 5], and will be enough to establish the sharpness of the bound in Theorem 2.7. However, since A_{17} is a smaller example, we opted to include that too.

We can now resolve the second conjecture of Dotsenko.

Corollary 2.5. *Let A satisfy the $t_4 = 0$ identity. It is not necessarily the case that $t_5 = 0$.*

Proof. In the algebra A_{17}, one computes $t_5(b_0, b_0, b_0, b_0, b_0) = 30b_4$. □

Remark 2.6. An alternative proof runs as follows; in [5], just before the proof of Theorem 5, it is shown that if $t_4 = t_5 = 0$ in some algebra A then the 5-Engel identity holds. As the 5-Engel identity does not hold for A_{17}, it is not possible for the $t_5 = 0$ identity to hold.

Next, we verify the second main result of this paper:

Theorem 2.7.

(1) Let A be an algebra satisfying the $t_3 = 0$ identity. Then A also satisfies the identity $e_3 = 0$.
(2) Let A be an algebra satisfying the $t_4 = 0$ identity. Then A also satisfies the identity $e_9 = 0$.

Proof. To prove the first claim, it is easily checked by hand that:

$$e_3(x, y) = \frac{1}{4}t_3(xy, y, y) - \frac{1}{4}t_3(y^2, x, y) + \frac{1}{4}[t_3(x, y, y) \cdot y] + \frac{1}{12}[t_3(y, y, y) \cdot x]$$

Thus if t_3 is identically zero then so is e_3.

The strategy behind the second claim is, in principle, exactly the same. However, the expression for $e_9(x, y)$ in terms of t_4 is significantly longer and cannot be checked by hand.

To do this, consider the free commutative non-associative algebra on two generators x and y. This has a basis given by binary trees. Consider the subspace $T_{1,9}$ consisting of monomials with

one occurrence of x and nine occurrences of y. We may consider the subspace of $T_{1,9}$ obtained by substituting all possible choices of monomials into t_4 and then multiplying by other monomials. Using this description along Gaussian elimination, we found an expression for $e_9(x,y)$ as a linear combination of 482 such terms, which is given in Appendix A.

Once one has the correct description, one can check it writing these 482 expressions in the tree basis as the columns of a 582×482 matrix M. In this system $e_9(x,y)$ is a basis element, and we have an explicit vector w such that $Mw = e_9(x,y)$.

This calculation is formalized in a Macaulay2 program `verify_e9.m2` in the accompanying GitHub repository [3]. This program expands the 482 terms in the tree basis and sums them appropriately. This constructs the matrix M, checks it and verifies that that all coefficients of Mw vanish except that of $e_9(x,y)$. □

As mentioned in the introduction, the form of Theorem 2.7 immediately suggests that if $k \geq 2$ and A satisfies the $t_k = 0$ identity then A should also satisfy the 3^{k-2}-Engel identity. We were able to give a lower bound of 14 via the following theorem, though clearly this is some way off 27.

Theorem 2.8. *There exists a 111-dimensional algebra A_{111} satisfying the $t_5 = 0$ identity for which there exists $a, b \in A_{111}$ with $e_{13}(a,b) \neq 0$.*

Proof. The proof follows the same lines as Theorems 2.1 and 2.3. The full multiplication table for the algebra can be found as `e13/multiplication_table.pdf` and the Macaulay2 verification as `e13/verify_e13.m2` in [3]. The latter takes about 10 minutes to run to completion on the author's machine. □

We conclude with a comment on the applicability of our results outside characteristic zero.

Remark 2.9. The structure coefficients in A_{17} and A_{22} are contained in $\mathbb{Z}\left[\frac{1}{4}\right]$. It follows that Theorems 2.1 and 2.3 are true for fields of characteristic greater than 2. Similarly, all of the coefficients in the appendix are contained in $\mathbb{Z}\left[\frac{1}{210}\right]$. It follows that Theorem 2.7 is true for fields of characteristic greater than 7.

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Appendices

A. Form of e_9 identity

In this appendix we give an expression for e_9 in terms of t_4. This identity was verified by Macaulay2. This is unlikely to be the shortest such expression.

$$\begin{aligned} e_9(x,y) = &\tfrac{81}{1000}\,[y\cdot[t_4(y,y^2,y^2,y(yx))\cdot y]] \\ &+\tfrac{293}{4500}\,[y\cdot[y\cdot[t_4(y,y,y,y(y(yx)))\cdot y]]] \\ &+\tfrac{43}{700}\,[t_4(y,y,y(yy^2),y(yx))\cdot y] \\ &+\tfrac{53}{875}\,[y\cdot[y\cdot[y\cdot[t_4(y,y,y,yx)\cdot y]]]] \\ &+\tfrac{83}{1400}\,[y\cdot[t_4(y,y,y(y(yy^2)),x)\cdot y]] \end{aligned}$$

$$
\begin{aligned}
&- \tfrac{41}{700}\,[t_4(y, y^2, yy^2, y(yx)) \cdot y] \\
&- \tfrac{7}{120}\,[t_4(y, y, y(y(yy^2)), yx) \cdot y] \\
&+ \tfrac{1529}{31500}\,[y \cdot [y \cdot [y \cdot [t_4(y, y, y, y(yx)) \cdot y]]]] \\
&+ \tfrac{487}{10500}\,[t_4(y, y, y^2, y(y(y(yx)))) \cdot y] \\
&+ \tfrac{1919}{42000}\,[y \cdot [t_4(y, y, y^2, y^2(yx)) \cdot y]] \\
&- \tfrac{9}{200}\,[y \cdot [t_4(y, y^2, y(yy^2), x) \cdot y]] \\
&- \tfrac{547}{12600}\,[y \cdot [t_4(y, y, y, y(y^2(yx))) \cdot y]] \\
&- \tfrac{149}{3500}\,t_4(y, y, y^2, y(y(y(y(yx))))) \\
&- \tfrac{89}{2100}\,[y \cdot [t_4(y, y, y, y^2(y(yx))) \cdot y]] \\
&- \tfrac{573}{14000}\,[y \cdot [y \cdot [t_4(y, y, y^2, yx) \cdot [y \cdot y]]]] \\
&- \tfrac{373}{10500}\,[y \cdot [y \cdot [t_4(y, y, yy^2, yx) \cdot y]]] \\
&+ \tfrac{341}{10500}\,[t_4(y, y, y^2, y(y^2(yx))) \cdot y] \\
&+ \tfrac{409}{12600}\,[y \cdot [y \cdot [[t_4(y, y, y, yx) \cdot y] \cdot [y \cdot y]]]] \\
&+ \tfrac{2033}{63000}\,[y \cdot [t_4(y, y, y, y(yx)) \cdot [y \cdot [y \cdot y]]]] \\
&+ \tfrac{29}{900}\,[y \cdot [t_4(y, y, y, y(yy^2)) \cdot [y \cdot x]]] \\
&- \tfrac{9}{280}\,t_4(y, y, y(y(y(yy^2))), yx) \\
&- \tfrac{223}{7000}\,[y \cdot [y \cdot [t_4(y, y^2, y^2, yx) \cdot y]]] \\
&+ \tfrac{31}{1050}\,t_4(y, y, y(yy^2), y(y(yx))) \\
&+ \tfrac{407}{14000}\,[y \cdot [y \cdot [y \cdot [t_4(y, y, y^2, x) \cdot [y \cdot y]]]]] \\
&- \tfrac{43}{1500}\,[y \cdot [y \cdot [y \cdot [[t_4(y, y, y, x) \cdot y] \cdot [y \cdot y]]]]] \\
&+ \tfrac{39}{1400}\,[y \cdot [y \cdot [t_4(y, y^2, yy^2, x) \cdot y]]] \\
&- \tfrac{113}{4200}\,[y \cdot [y \cdot [y \cdot [t_4(y, y, y^2, yx) \cdot y]]]] \\
&- \tfrac{2}{75}\,[y \cdot [y \cdot [t_4(y, y, y, y) \cdot [y \cdot [y \cdot [y \cdot x]]]]]] \\
&+ \tfrac{19}{750}\,[y \cdot [t_4(y, y, y, (y(yy^2))x) \cdot y]] \\
&- \tfrac{113}{4500}\,[t_4(y, y, y, y(y(y(y(yx))))) \cdot y] \\
&+ \tfrac{1}{40}\,t_4(y, y^2, y(y(y(yy^2))), x) \\
&+ \tfrac{87}{3500}\,[y \cdot [y \cdot [y \cdot [t_4(y, y^2, y^2, x) \cdot y]]]] \\
&- \tfrac{62}{2625}\,[y \cdot [t_4(y, y, y, y(y(y(yx)))) \cdot y]] \\
&- \tfrac{33}{1400}\,[y \cdot [y \cdot [t_4(y, y, y^2, y(yx)) \cdot y]]] \\
&- \tfrac{7}{300}\,t_4(y, y^2, yy^2, y(y(yx))) \\
&- \tfrac{97}{4200}\,[y \cdot [t_4(y, y^2, yy^2, yx) \cdot y]] \\
&- \tfrac{479}{21000}\,[y \cdot [y \cdot [y \cdot [y \cdot [t_4(y, y, y, x) \cdot [y \cdot y]]]]]] \\
&+ \tfrac{67}{3000}\,[y \cdot [t_4(y, y, yy^2, y(yx)) \cdot y]] \\
&+ \tfrac{701}{31500}\,[y \cdot [y \cdot [t_4(y, y, y, y^2(yx)) \cdot y]]] \\
&+ \tfrac{1}{45}\,t_4(y, y, y, y(y(y(y(y(yx)))))) \\
&- \tfrac{1}{48}\,[t_4(y, y, y^2, y(y(yy^2))) \cdot x] \\
&- \tfrac{257}{12600}\,[y \cdot [y \cdot [t_4(y, y, y, y(yy^2)) \cdot x]]] \\
&- \tfrac{271}{14000}\,[t_4(y, y^2, y^2, y^2(yx)) \cdot y]
\end{aligned}
$$

$$
\begin{aligned}
&- \tfrac{27}{1400}\,[y\cdot[y\cdot[y\cdot[y\cdot[t_4(y,y,y^2,x)\cdot y]]]]] \\
&+ \tfrac{2}{105}\,t_4(y,y,y(yy^2),y^2(yx)) \\
&+ \tfrac{2}{105}\,t_4(y,y,y(y(yy^2)),y(yx)) \\
&+ \tfrac{199}{10500}\,[y\cdot[y\cdot[y\cdot[y\cdot[y\cdot[t_4(y,y,y,x)\cdot y]]]]]] \\
&- \tfrac{113}{6000}\,[y\cdot[[y\cdot y]\cdot[t_4(y,y,y,yx)\cdot[y\cdot y]]]] \\
&+ \tfrac{169}{9000}\,[y\cdot[t_4(y,y,y,y^2)\cdot[y\cdot[y\cdot[y\cdot x]]]]] \\
&- \tfrac{7}{375}\,[y\cdot[t_4(y,y,y,y)\cdot[y\cdot[y\cdot[y\cdot[y\cdot x]]]]]] \\
&- \tfrac{13}{700}\,[y\cdot[y\cdot[y\cdot[t_4(y,y,y,x)\cdot[y\cdot[y\cdot y]]]]]] \\
&+ \tfrac{97}{5250}\,t_4(y,y^2,y^2,y(y(y(yx)))) \\
&- \tfrac{1}{56}\,[t_4(y,y,y(y(y(yy^2))),x)\cdot y] \\
&- \tfrac{77}{4500}\,[y\cdot[y\cdot[t_4(y,y,y,y^2)\cdot[y\cdot[y\cdot x]]]]] \\
&+ \tfrac{1}{60}\,t_4(y,y,y(y(y(y(yy^2)))),x) \\
&+ \tfrac{33}{2000}\,[y\cdot[t_4(y,y^2,y^2,yx)\cdot[y\cdot y]]] \\
&+ \tfrac{229}{14000}\,[y\cdot[t_4(y,y,yy^2,yx)\cdot[y\cdot y]]] \\
&- \tfrac{997}{63000}\,[y\cdot[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot[y\cdot[y\cdot x]]]]] \\
&+ \tfrac{124}{7875}\,[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y^2)\cdot[y\cdot x]]]]] \\
&+ \tfrac{31}{2000}\,[y\cdot[y\cdot[t_4(y,y,y^2,x)\cdot[y\cdot[y\cdot y]]]]] \\
&+ \tfrac{27}{1750}\,[y\cdot[t_4(y,y,y,yy^2)\cdot[y\cdot[y\cdot x]]]] \\
&+ \tfrac{3}{200}\,[y\cdot[y\cdot[t_4(y,y,y,x)\cdot[y\cdot[y\cdot[y\cdot y]]]]]] \\
&+ \tfrac{3}{200}\,[y\cdot[t_4(y,y,y(yy^2),x)\cdot[y\cdot y]]] \\
&- \tfrac{227}{15750}\,t_4(y,y,y,y^2(y(y(y(yx))))) \\
&+ \tfrac{47}{3500}\,[y\cdot[y\cdot[t_4(y,y,y,y(y^2x))\cdot y]]] \\
&- \tfrac{137}{10500}\,[t_4(y,y,yy^2,y(y(yx)))\cdot y] \\
&+ \tfrac{179}{14000}\,[y\cdot[t_4(y,y,y^2,y(yx))\cdot[y\cdot y]]] \\
&+ \tfrac{1}{80}\,[[y\cdot[t_4(y,y,y^2,yy^2)\cdot y]]\cdot x] \\
&+ \tfrac{521}{42000}\,[y\cdot[y\cdot[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot[y\cdot x]]]]] \\
&- \tfrac{43}{3500}\,t_4(y,y,yy^2,y(y(y(yx)))) \\
&+ \tfrac{109}{9000}\,[y\cdot[y\cdot[y\cdot[t_4(y,y,y,yx)\cdot[y\cdot y]]]]] \\
&- \tfrac{33}{2800}\,[y\cdot[y\cdot[t_4(y,y,y^2,y^2x)\cdot y]]] \\
&- \tfrac{367}{31500}\,[t_4(y,y,y,y^2(y(y(yx))))\cdot y] \\
&- \tfrac{479}{42000}\,[y\cdot[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y^2)\cdot x]]]]] \\
&+ \tfrac{89}{7875}\,[y\cdot[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y)\cdot[y\cdot x]]]]]] \\
&- \tfrac{47}{4200}\,t_4(y,y,y,(y(y(y(yy^2))))x) \\
&+ \tfrac{39}{3500}\,[y\cdot[t_4(y,y,y^2,y(y^2x))\cdot y]] \\
&+ \tfrac{173}{15750}\,[y\cdot[t_4(y,y,y,y(y(yx)))\cdot[y\cdot y]]] \\
&- \tfrac{229}{21000}\,[y\cdot[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,yx)\cdot y]]]] \\
&- \tfrac{113}{10500}\,[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y^2x)\cdot y]]]] \\
&- \tfrac{19}{1800}\,[[y\cdot[y\cdot[t_4(y,y,y,yy^2)\cdot y]]]\cdot x]
\end{aligned}
$$

$$
\begin{aligned}
&- \tfrac{31}{3000}\,[y\cdot[y\cdot[[t_4(y,y,y,x)\cdot y]\cdot[y\cdot[y\cdot y]]]]]\\
&+ \tfrac{43}{4200}\,[y\cdot[y\cdot[t_4(y,y,y,yy^2)\cdot[y\cdot x]]]]\\
&+ \tfrac{23}{2250}\,[t_4(y,y,y,y(y(yx)))\cdot[y\cdot[y\cdot y]]]\\
&- \tfrac{1}{100}\,[t_4(y,y,y(yy^2),y^2x)\cdot y]\\
&+ \tfrac{157}{15750}\,[[t_4(y,y,y,y(y(yx)))\cdot y]\cdot[y\cdot y]]\\
&- \tfrac{89}{9000}\,[y\cdot[y\cdot[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot[y\cdot x]]]]]\\
&- \tfrac{41}{4200}\,[y\cdot[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot[y\cdot[y\cdot x]]]]]\\
&- \tfrac{407}{42000}\,[y\cdot[y\cdot[y\cdot[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot x]]]]]\\
&+ \tfrac{101}{10500}\,t_4(y,y,y,y(y(y^2(y(yx)))))\\
&- \tfrac{149}{15750}\,[y\cdot[y\cdot[t_4(y,y,y,y(yx))\cdot[y\cdot y]]]]\\
&+ \tfrac{59}{6300}\,[y\cdot[y\cdot[[t_4(y,y,y,y)\cdot[y\cdot[y\cdot y]]]\cdot x]]]\\
&+ \tfrac{59}{6300}\,[y\cdot[y\cdot[[t_4(y,y,y,y^2)\cdot[y\cdot y]]\cdot x]]]\\
&- \tfrac{29}{3150}\,[y\cdot[[t_4(y,y,y,yx)\cdot y]\cdot[y\cdot[y\cdot y]]]]\\
&- \tfrac{8}{875}\,[y\cdot[y\cdot[y\cdot[t_4(y,y,yy^2,x)\cdot y]]]]\\
&- \tfrac{191}{21000}\,t_4(y,y,y^2,y(y^2(y(yx))))\\
&- \tfrac{19}{2100}\,t_4(y,y,y,y(y^2(y^2(yx))))\\
&- \tfrac{2}{225}\,[y\cdot[[t_4(y,y,y,y)\cdot[y\cdot[y\cdot y]]]\cdot[y\cdot x]]]\\
&- \tfrac{2}{225}\,[y\cdot[[t_4(y,y,y,y^2)\cdot[y\cdot y]]\cdot[y\cdot x]]]\\
&- \tfrac{31}{3500}\,[t_4(y,y^2,y^2,y(y(yx)))\cdot y]\\
&+ \tfrac{223}{25200}\,[y\cdot[y\cdot[[y\cdot[t_4(y,y,y,y)\cdot[y\cdot y]]]\cdot x]]]\\
&- \tfrac{223}{25200}\,[y\cdot[y\cdot[t_4(y,y^2,y^2,y^2)\cdot x]]]\\
&+ \tfrac{37}{4200}\,[y\cdot[t_4(y,y,y^2,yx)\cdot[y\cdot[y\cdot y]]]]\\
&- \tfrac{79}{9000}\,[y\cdot[y\cdot[t_4(y,y,y,y)\cdot[[y\cdot y]\cdot[y\cdot x]]]]]\\
&+ \tfrac{13}{1500}\,t_4(y,y,y,(y(yy^2))(y(yx)))\\
&- \tfrac{3}{350}\,[[y\cdot y]\cdot[t_4(y,y,y^2,yx)\cdot[y\cdot y]]]\\
&- \tfrac{179}{21000}\,[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y)\cdot[y\cdot[y\cdot x]]]]]]\\
&+ \tfrac{47}{5600}\,t_4(y,y,y^2,y^2(y^2(yx)))\\
&- \tfrac{117}{14000}\,[t_4(y,y,y^2,y^2(y^2x))\cdot y]\\
&- \tfrac{1}{120}\,[t_4(y,y,y,x)\cdot[y\cdot[y\cdot[y\cdot[y\cdot[y\cdot y]]]]]]\\
&+ \tfrac{1}{120}\,[[t_4(y,y,y^2,yy^2)\cdot[y\cdot y]]\cdot x]\\
&+ \tfrac{1}{120}\,[[t_4(y,y,yy^2,yy^2)\cdot y]\cdot x]\\
&- \tfrac{1}{120}\,[t_4(y,y,yy^2,y(yy^2))\cdot x]\\
&- \tfrac{1}{120}\,[t_4(y,y,y(yy^2),yx)\cdot[y\cdot y]]\\
&+ \tfrac{517}{63000}\,[y\cdot[t_4(y,y,y,y^2)\cdot[[y\cdot y]\cdot[y\cdot x]]]]\\
&- \tfrac{257}{31500}\,t_4(y,y,y,y(y^2(y(y(yx)))))\\
&+ \tfrac{17}{2100}\,t_4(y,y^2,y^2,y(y^2(yx)))\\
&- \tfrac{29}{3600}\,[y\cdot[[y\cdot[t_4(y,y,y,y)\cdot[y\cdot y]]]\cdot[y\cdot x]]]\\
&+ \tfrac{29}{3600}\,[y\cdot[t_4(y,y^2,y^2,y^2)\cdot[y\cdot x]]]
\end{aligned}
$$

$$+ \tfrac{253}{31500}\,[t_4(y,y,y,y)\cdot[y\cdot[y\cdot[y\cdot[y\cdot x]]]]]$$
$$- \tfrac{83}{10500}\,[t_4(y,y,y,y((y(yy^2))x))\cdot y]$$
$$+ \tfrac{11}{1400}\,[y\cdot[t_4(y,y,y^2,y(y(yx)))\cdot y]]$$
$$- \tfrac{11}{1400}\,[t_4(y,y^2,y^2,y(y^2x))\cdot y]$$
$$- \tfrac{107}{14000}\,[y\cdot[t_4(y,y^2,y^2,y^2x)\cdot y]]$$
$$- \tfrac{47}{6300}\,[t_4(y,y,y,y(y(y^2(yx))))\cdot y]$$
$$- \tfrac{11}{1500}\,[y\cdot[[y\cdot y]\cdot[t_4(y,y,y,y^2)\cdot[y\cdot x]]]]$$
$$- \tfrac{61}{8400}\,[t_4(y,y,y,(y(y(yy^2)))x)\cdot y]$$
$$- \tfrac{99}{14000}\,[y\cdot[t_4(y,y,y,y)\cdot[y\cdot[[y\cdot y]\cdot[y\cdot x]]]]]$$
$$- \tfrac{99}{14000}\,[y\cdot[t_4(y,y^2,y^2,x)\cdot[y\cdot[y\cdot y]]]]$$
$$- \tfrac{11}{1575}\,[t_4(y,y,y,y(yy^2))\cdot[[y\cdot y]\cdot x]]$$
$$+ \tfrac{11}{1575}\,[[y\cdot y]\cdot[[t_4(y,y,y,yx)\cdot y]\cdot[y\cdot y]]]$$
$$+ \tfrac{1}{144}\,[[t_4(y,y,y,y)\cdot[y\cdot[y\cdot[y\cdot[y\cdot y]]]]]\cdot x]$$
$$+ \tfrac{29}{4200}\,[y\cdot[t_4(y,y,y(yy^2),yx)\cdot y]]$$
$$- \tfrac{71}{10500}\,t_4(y,y,yy^2,y^2(y(yx)))$$
$$- \tfrac{11}{1680}\,t_4(y,y^2,yy^2,y^2(yx))$$
$$+ \tfrac{17}{2625}\,[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y)\cdot[[y\cdot y]\cdot x]]]]]$$
$$- \tfrac{9}{1400}\,[y\cdot[y\cdot[t_4(y,y,y(yy^2),x)\cdot y]]]$$
$$- \tfrac{11}{1750}\,[y\cdot[t_4(y,y,yy^2,y^2x)\cdot y]]$$
$$- \tfrac{79}{12600}\,t_4(y,y,y,y^2(y(y^2(yx))))$$
$$+ \tfrac{1}{160}\,[y\cdot[[y\cdot y]\cdot[t_4(y,y,y^2,x)\cdot[y\cdot y]]]]$$
$$- \tfrac{43}{7000}\,[y\cdot[[t_4(y,y,y,y)\cdot[y\cdot y]]\cdot[y\cdot[y\cdot x]]]]$$
$$+ \tfrac{43}{7000}\,[y\cdot[t_4(y,y,y^2,y^2)\cdot[y\cdot[y\cdot x]]]]$$
$$- \tfrac{187}{31500}\,[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot[y\cdot[y\cdot[y\cdot x]]]]]$$
$$+ \tfrac{37}{6300}\,t_4(y,y,y,(y(y(yy^2)))(yx))$$
$$- \tfrac{7}{1200}\,t_4(y,y,y^2,y(y(y^2(yx))))$$
$$+ \tfrac{61}{10500}\,[t_4(y,y,y,yy^2)\cdot[y\cdot[y\cdot[y\cdot x]]]]$$
$$- \tfrac{73}{12600}\,[[y\cdot y]\cdot[t_4(y,y,y,y(yy^2))\cdot x]]$$
$$- \tfrac{121}{21000}\,[y\cdot[y\cdot[y\cdot[t_4(y,y,y,yy^2)\cdot x]]]]$$
$$- \tfrac{1}{180}\,[[[t_4(y,y,y,yy^2)\cdot y]\cdot[y\cdot y]]\cdot x]$$
$$- \tfrac{1}{180}\,[[y\cdot[t_4(y,y,y,y(yy^2))\cdot y]]\cdot x]$$
$$+ \tfrac{1}{180}\,[t_4(y,y,y,y(y(y(yy^2))))\cdot x]$$
$$+ \tfrac{1}{180}\,t_4(y,y,y,y((y(yy^2))(yx)))$$
$$+ \tfrac{233}{42000}\,[t_4(y,y,y,y(y^2(y^2x)))\cdot y]$$
$$- \tfrac{11}{2000}\,[y\cdot[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,y)\cdot[y\cdot x]]]]]$$
$$+ \tfrac{17}{3150}\,[t_4(y,y,y,y^2(y^2(yx)))\cdot y]$$
$$- \tfrac{3}{560}\,[y\cdot[t_4(y,y^2,yy^2,x)\cdot[y\cdot y]]]$$
$$+ \tfrac{37}{7000}\,[y\cdot[y\cdot[y\cdot[[t_4(y,y,y,y)\cdot[y\cdot y]]\cdot x]]]]$$

$$
\begin{aligned}
&- \tfrac{37}{7000}\,[y\cdot[y\cdot[y\cdot[t_4(y,y,y^2,y^2)\cdot x]]]]\\
&- \tfrac{31}{6000}\,[y\cdot[y\cdot[[y\cdot[y\cdot y]]\cdot[t_4(y,y,y,y)\cdot x]]]]\\
&- \tfrac{323}{63000}\,[y\cdot[y\cdot[y\cdot[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot x]]]]]\\
&- \tfrac{43}{8400}\,t_4(y,y,yy^2,y(y^2(yx)))\\
&+ \tfrac{319}{63000}\,[y\cdot[y\cdot[t_4(y,y,y,yx)\cdot[y\cdot[y\cdot y]]]]]\\
&+ \tfrac{1}{200}\,[[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y)\cdot[y\cdot y]]]]]\cdot x]\\
&- \tfrac{1}{200}\,[[y\cdot[y\cdot[t_4(y,y,y^2,y^2)\cdot y]]]\cdot x]\\
&- \tfrac{1}{200}\,[t_4(y,y^2,y(yy^2),x)\cdot[y\cdot y]]\\
&- \tfrac{13}{2625}\,[y\cdot[t_4(y,y,y,y(y(y^2x)))\cdot y]]\\
&- \tfrac{103}{21000}\,[t_4(y,y,y^2,y(y(yx)))\cdot[y\cdot y]]\\
&- \tfrac{281}{63000}\,[t_4(y,y,y,(y(yy^2))(yx))\cdot y]\\
&+ \tfrac{1}{225}\,[[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y^2)\cdot y]]]]\cdot x]\\
&+ \tfrac{137}{31500}\,[[y\cdot y]\cdot[t_4(y,y,y,y(yx))\cdot[y\cdot y]]]\\
&- \tfrac{3}{700}\,t_4(y,y,y(yy^2),y(y^2x))\\
&+ \tfrac{179}{42000}\,[y\cdot[[t_4(y,y,y,x)\cdot[y\cdot y]]\cdot[y\cdot[y\cdot y]]]]\\
&+ \tfrac{67}{15750}\,t_4(y,y,y,y(y(y(y^2(yx)))))\\
&- \tfrac{1}{240}\,[[y\cdot[y\cdot[t_4(y,y,y,y)\cdot[y\cdot[y\cdot y]]]]]\cdot x]\\
&+ \tfrac{1}{240}\,[[y\cdot[y\cdot[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot y]]]]\cdot x]\\
&- \tfrac{1}{240}\,[[y\cdot[y\cdot[t_4(y,y,y,y^2)\cdot[y\cdot y]]]]\cdot x]\\
&- \tfrac{29}{7000}\,t_4(y,y,y^2,y(y(y(y^2x))))\\
&- \tfrac{43}{10500}\,[t_4(y,y,y^2,y^2(y(yx)))\cdot y]\\
&- \tfrac{167}{42000}\,[t_4(y,y,yy^2,y^2(yx))\cdot y]\\
&+ \tfrac{2}{525}\,[y\cdot[y\cdot[[t_4(y,y,y,y)\cdot[y\cdot y]]\cdot[y\cdot x]]]]\\
&- \tfrac{2}{525}\,[y\cdot[y\cdot[t_4(y,y,y^2,y^2)\cdot[y\cdot x]]]]\\
&+ \tfrac{2}{525}\,t_4(y,y,y^2,y^2(y(y(yx))))\\
&+ \tfrac{2}{525}\,[t_4(y,y^2,y^2,yx)\cdot[y\cdot[y\cdot y]]]\\
&- \tfrac{2}{525}\,[t_4(y,y^2,yy^2,yx)\cdot[y\cdot y]]\\
&+ \tfrac{53}{14000}\,[y\cdot[y\cdot[t_4(y,y,yy^2,x)\cdot[y\cdot y]]]]\\
&- \tfrac{157}{42000}\,[y\cdot[y\cdot[[y\cdot y]\cdot[t_4(y,y,y,x)\cdot[y\cdot y]]]]]\\
&- \tfrac{13}{3500}\,[t_4(y,y,y,y(y(y(y^2x))))\cdot y]\\
&+ \tfrac{13}{3500}\,[y\cdot[y\cdot[t_4(y,y^2,y^2,x)\cdot[y\cdot y]]]]\\
&+ \tfrac{51}{14000}\,[y\cdot[[t_4(y,y,y^2,x)\cdot y]\cdot[y\cdot[y\cdot y]]]]\\
&- \tfrac{19}{5250}\,[[t_4(y,y,y,yx)\cdot y]\cdot[y\cdot[y\cdot[y\cdot y]]]]\\
&+ \tfrac{1}{280}\,[y\cdot[[y\cdot y]\cdot[t_4(y,y,y,x)\cdot[y\cdot[y\cdot y]]]]]\\
&+ \tfrac{1}{280}\,t_4(y,y,y(y(yy^2)),y^2x)\\
&- \tfrac{1}{280}\,[t_4(y,y^2,yy^2,x)\cdot[y\cdot[y\cdot y]]]\\
&- \tfrac{11}{3150}\,[t_4(y,y,y,y(yy^2))\cdot[y\cdot[y\cdot x]]]\\
&+ \tfrac{73}{21000}\,[t_4(y,y,y^2,y(yx))\cdot[y\cdot[y\cdot y]]]
\end{aligned}
$$

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\begin{aligned}
&- \tfrac{29}{8400}\,[t_4(y,y,y^2,yy^2)\cdot[y\cdot[y\cdot x]]] \\
&- \tfrac{1}{300}\,[y\cdot[[y\cdot[t_4(y,y,y,x)\cdot y]]\cdot[y\cdot[y\cdot y]]]] \\
&+ \tfrac{1}{300}\,[[t_4(y,y,y,y^2(yx))\cdot y]\cdot[y\cdot y]] \\
&- \tfrac{1}{300}\,[t_4(y,y,y,y(y(y(yx))))\cdot[y\cdot y]] \\
&+ \tfrac{17}{5250}\,[t_4(y,y,y,y)\cdot[y\cdot[[y\cdot y]\cdot[y\cdot[y\cdot x]]]]] \\
&+ \tfrac{17}{5250}\,[y\cdot[y\cdot[t_4(y,y,y,y^2)\cdot[[y\cdot y]\cdot x]]]] \\
&+ \tfrac{11}{3500}\,[t_4(y,y^2,y^2,y(yx))\cdot[y\cdot y]] \\
&+ \tfrac{131}{42000}\,[y\cdot[y\cdot[y\cdot[y\cdot[[t_4(y,y,y,y)\cdot y]\cdot x]]]]] \\
&- \tfrac{31}{10500}\,[[t_4(y,y,y,y)\cdot[y\cdot y]]\cdot[y\cdot[y\cdot[y\cdot x]]]] \\
&+ \tfrac{31}{10500}\,t_4(y,y,y,y(y(y(y(y^2x)))))\\
&+ \tfrac{31}{10500}\,[t_4(y,y,y^2,y^2)\cdot[y\cdot[y\cdot[y\cdot x]]]] \\
&- \tfrac{73}{25200}\,[t_4(y,y,y,y(y^2(yx)))\cdot[y\cdot y]] \\
&+ \tfrac{1}{350}\,[[y\cdot y]\cdot[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot[y\cdot x]]]] \\
&+ \tfrac{17}{6000}\,[y\cdot[t_4(y,y,y,y^2x)\cdot[y\cdot[y\cdot y]]]] \\
&- \tfrac{47}{16800}\,[t_4(y,y,y,y)\cdot[[y\cdot y]\cdot[[y\cdot y]\cdot[y\cdot x]]]] \\
&+ \tfrac{1}{360}\,[[y\cdot[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot[y\cdot y]]]]\cdot x] \\
&- \tfrac{1}{360}\,[[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot[y\cdot[y\cdot y]]]]\cdot x] \\
&+ \tfrac{1}{360}\,[[[y\cdot y]\cdot[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot y]]]\cdot x] \\
&- \tfrac{1}{360}\,[[y\cdot[t_4(y,y,y,y^2)\cdot[y\cdot[y\cdot y]]]]\cdot x] \\
&- \tfrac{1}{360}\,[[[y\cdot y]\cdot[t_4(y,y,y,y^2)\cdot[y\cdot y]]]\cdot x] \\
&+ \tfrac{1}{360}\,[[t_4(y,y,y,y(yy^2))\cdot[y\cdot y]]\cdot x] \\
&+ \tfrac{1}{360}\,[[t_4(y,y,y,y(y(yy^2)))\cdot y]\cdot x] \\
&- \tfrac{1}{360}\,[t_4(y,y,y,y^2(y(yy^2)))\cdot x] \\
&+ \tfrac{23}{8400}\,[y\cdot[y\cdot[t_4(y,y,y,y^2x)\cdot[y\cdot y]]]] \\
&+ \tfrac{17}{6300}\,[y\cdot[[t_4(y,y,y,y(yy^2))\cdot y]\cdot x]] \\
&- \tfrac{11}{4200}\,[t_4(y,y,yy^2,yx)\cdot[y\cdot[y\cdot y]]] \\
&+ \tfrac{4}{1575}\,[[y\cdot y]\cdot[t_4(y,y,y,yx)\cdot[y\cdot[y\cdot y]]]] \\
&- \tfrac{53}{21000}\,[y\cdot[y\cdot[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,x)\cdot y]]]]] \\
&+ \tfrac{1}{400}\,[y\cdot[t_4(y,y,y^2,yy^2)\cdot[y\cdot x]]] \\
&+ \tfrac{41}{16800}\,[[y\cdot y]\cdot[t_4(y,y,y^2,yy^2)\cdot x]] \\
&- \tfrac{17}{7000}\,[[y\cdot[y\cdot y]]\cdot[t_4(y,y,y,y)\cdot[y\cdot[y\cdot x]]]] \\
&+ \tfrac{17}{7000}\,[[t_4(y,y,y^2,x)\cdot y]\cdot[y\cdot[y\cdot[y\cdot y]]]] \\
&+ \tfrac{17}{7000}\,[t_4(y,y,y^2,yx)\cdot[y\cdot[y\cdot[y\cdot y]]]] \\
&+ \tfrac{17}{7000}\,t_4(y,y^2,y^2,y^2(y^2x)) \\
&- \tfrac{1}{420}\,[y\cdot[[y\cdot y]\cdot[[t_4(y,y,y,x)\cdot y]\cdot[y\cdot y]]]] \\
&+ \tfrac{1}{420}\,[[y\cdot[y\cdot y]]\cdot[t_4(y,y,y,x)\cdot[y\cdot[y\cdot y]]]] \\
&+ \tfrac{29}{12600}\,[[t_4(y,y,y,yy^2)\cdot y]\cdot[y\cdot[y\cdot x]]] \\
&- \tfrac{2}{875}\,t_4(y,y,y^2,y(y^2(y^2x)))
\end{aligned}
$$

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\begin{aligned}
&- \tfrac{19}{8400}\,[[t_4(y,y,y^2,yy^2)\cdot y]\cdot[y\cdot x]]\\
&- \tfrac{71}{31500}\,[[t_4(y,y,y,y(yx))\cdot y]\cdot[y\cdot[y\cdot y]]]\\
&+ \tfrac{17}{7875}\,[[y\cdot y]\cdot[t_4(y,y,y,yy^2)\cdot[y\cdot x]]]\\
&+ \tfrac{3}{1400}\,[t_4(y,y,y,y^2(y(y^2x)))\cdot y]\\
&- \tfrac{3}{1400}\,t_4(y,y,y,y((y(y(yy^2)))x))\\
&+ \tfrac{3}{1400}\,[t_4(y,y^2,yy^2,y^2x)\cdot y]\\
&+ \tfrac{179}{84000}\,[y\cdot[[y\cdot[y\cdot y]]\cdot[t_4(y,y,y,y^2)\cdot x]]]\\
&- \tfrac{11}{5250}\,t_4(y,y,y,y(y((y(yy^2))x)))\\
&- \tfrac{1}{480}\,[y\cdot[[y\cdot y]\cdot[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot x]]]]\\
&- \tfrac{43}{21000}\,[y\cdot[[y\cdot[t_4(y,y,y,y)\cdot y]]\cdot[y\cdot[y\cdot x]]]]\\
&- \tfrac{43}{21000}\,[y\cdot[[t_4(y,y,y,y^2)\cdot y]\cdot[y\cdot[y\cdot x]]]]\\
&+ \tfrac{17}{8400}\,[[t_4(y,y,y,y)\cdot[y\cdot[y\cdot y]]]\cdot[y\cdot[y\cdot x]]]\\
&+ \tfrac{17}{8400}\,[[t_4(y,y,y,y^2)\cdot[y\cdot y]]\cdot[y\cdot[y\cdot x]]]\\
&- \tfrac{1}{500}\,[[y\cdot[t_4(y,y,y,x)\cdot y]]\cdot[y\cdot[y\cdot[y\cdot y]]]]\\
&- \tfrac{1}{500}\,[t_4(y,y,y,y(yx))\cdot[y\cdot[y\cdot[y\cdot y]]]]\\
&+ \tfrac{83}{42000}\,[y\cdot[t_4(y,y,y,y)\cdot[[y\cdot y]\cdot[[y\cdot y]\cdot x]]]]\\
&+ \tfrac{41}{21000}\,[y\cdot[t_4(y,y,y,y)\cdot[y\cdot[y\cdot[[y\cdot y]\cdot x]]]]]\\
&- \tfrac{7}{3600}\,[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot[[y\cdot y]\cdot[y\cdot x]]]]\\
&+ \tfrac{7}{3600}\,[[y\cdot[[t_4(y,y,y,y^2)\cdot y]\cdot[y\cdot y]]]\cdot x]\\
&- \tfrac{1}{525}\,[[t_4(y,y,y,y)\cdot[y\cdot y]]\cdot[[y\cdot y]\cdot[y\cdot x]]]\\
&+ \tfrac{1}{525}\,[y\cdot[t_4(y,y,y,y^2)\cdot[y\cdot[[y\cdot y]\cdot x]]]]\\
&+ \tfrac{1}{525}\,[t_4(y,y,y,y(y(yy^2)))\cdot[y\cdot x]]\\
&- \tfrac{1}{525}\,[y\cdot[t_4(y,y,y,y(y(yy^2)))\cdot x]]\\
&+ \tfrac{1}{525}\,[t_4(y,y,y^2,y^2)\cdot[[y\cdot y]\cdot[y\cdot x]]]\\
&- \tfrac{157}{84000}\,[y\cdot[y\cdot[[y\cdot y]\cdot[t_4(y,y,y,y^2)\cdot x]]]]\\
&+ \tfrac{47}{25200}\,[[y\cdot y]\cdot[[y\cdot[t_4(y,y,y,y)\cdot[y\cdot y]]]\cdot x]]\\
&- \tfrac{47}{25200}\,[[y\cdot y]\cdot[t_4(y,y^2,y^2,y^2)\cdot x]]\\
&- \tfrac{13}{7000}\,[y\cdot[y\cdot[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y)\cdot x]]]]]]\\
&+ \tfrac{13}{7000}\,[[y\cdot y]\cdot[y\cdot[y\cdot[t_4(y,y,y^2,x)\cdot y]]]]\\
&+ \tfrac{11}{6000}\,[y\cdot[[y\cdot y]\cdot[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot x]]]]\\
&- \tfrac{23}{12600}\,[y\cdot[t_4(y,y,y,y^2(yx))\cdot[y\cdot y]]]\\
&+ \tfrac{1}{560}\,[y\cdot[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,x)\cdot[y\cdot y]]]]]\\
&+ \tfrac{1}{560}\,t_4(y,y,y^2,(y(y(yy^2)))x)\\
&+ \tfrac{37}{21000}\,[y\cdot[y\cdot[y\cdot[[y\cdot[t_4(y,y,y,y)\cdot y]]\cdot x]]]]\\
&+ \tfrac{37}{21000}\,[y\cdot[y\cdot[y\cdot[[t_4(y,y,y,y^2)\cdot y]\cdot x]]]]\\
&+ \tfrac{37}{21000}\,[[t_4(y,y,y,y^2x)\cdot y]\cdot[y\cdot[y\cdot y]]]\\
&- \tfrac{221}{126000}\,[y\cdot[[t_4(y,y,y,y)\cdot y]\cdot[[y\cdot y]\cdot[y\cdot x]]]]\\
&+ \tfrac{11}{6300}\,[[y\cdot[t_4(y,y,y,y)\cdot[y\cdot y]]]\cdot[[y\cdot y]\cdot x]]
\end{aligned}
$$

$$
\begin{aligned}
&+ \tfrac{11}{6300}\,[[t_4(y,y,y,y(yy^2))\cdot y]\cdot[y\cdot x]]\\
&- \tfrac{11}{6300}\,[t_4(y,y^2,y^2,y^2)\cdot[[y\cdot y]\cdot x]]\\
&+ \tfrac{29}{16800}\,[t_4(y,y,y^2,yy^2)\cdot[[y\cdot y]\cdot x]]\\
&- \tfrac{3}{1750}\,t_4(y,y,y,y^2(y^2(y(yx))))\\
&+ \tfrac{107}{63000}\,[[y\cdot[t_4(y,y,y,yy^2)\cdot y]]\cdot[y\cdot x]]\\
&+ \tfrac{1}{600}\,[[y\cdot[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot[y\cdot y]]]]\cdot x]\\
&+ \tfrac{1}{600}\,[[[t_4(y,y,y,y)\cdot[y\cdot y]]\cdot[y\cdot[y\cdot y]]]\cdot x]\\
&- \tfrac{1}{600}\,[y\cdot[[t_4(y,y,y,yy^2)\cdot y]\cdot[y\cdot x]]]\\
&- \tfrac{1}{600}\,[[y\cdot[t_4(y,y,y,yy^2)\cdot[y\cdot y]]]\cdot x]\\
&+ \tfrac{1}{600}\,[[y\cdot y]\cdot[t_4(y,y,y,x)\cdot[y\cdot[y\cdot[y\cdot y]]]]]\\
&+ \tfrac{1}{600}\,[t_4(y,y,y,(y(yy^2))x)\cdot[y\cdot y]]\\
&- \tfrac{1}{600}\,[[t_4(y,y,y^2,y^2)\cdot[y\cdot[y\cdot y]]]\cdot x]\\
&- \tfrac{1}{600}\,[[y\cdot[t_4(y,y,y^2,y^2)\cdot[y\cdot y]]]\cdot x]\\
&- \tfrac{41}{25200}\,[[y\cdot y]\cdot[[t_4(y,y,y,yy^2)\cdot y]\cdot x]]\\
&+ \tfrac{17}{10500}\,[t_4(y,y,yy^2,y(yx))\cdot[y\cdot y]]\\
&+ \tfrac{11}{7000}\,[y\cdot[t_4(y,y,y,y)\cdot[[y\cdot y]\cdot[y\cdot[y\cdot x]]]]]\\
&+ \tfrac{11}{7000}\,[t_4(y,y,y^2,y(y(y^2x)))\cdot y]\\
&- \tfrac{13}{8400}\,[y\cdot[y\cdot[t_4(y,y,y^2,yy^2)\cdot x]]]\\
&- \tfrac{4}{2625}\,[t_4(y,y,y,y(y^2x))\cdot[y\cdot[y\cdot y]]]\\
&+ \tfrac{19}{12600}\,[t_4(y,y,y,y^2(yx))\cdot[y\cdot[y\cdot y]]]\\
&- \tfrac{3}{2000}\,[t_4(y,y^2,y^2,y^2x)\cdot[y\cdot y]]\\
&- \tfrac{47}{31500}\,[[y\cdot[y\cdot y]]\cdot[y\cdot[t_4(y,y,y,y)\cdot[y\cdot x]]]]\\
&+ \tfrac{23}{15750}\,[t_4(y,y,y,y(y^2(y(yx))))\cdot y]\\
&+ \tfrac{61}{42000}\,[y\cdot[y\cdot[t_4(y,y,y,y)\cdot[y\cdot[[y\cdot y]\cdot x]]]]]\\
&- \tfrac{1}{700}\,[y\cdot[[y\cdot y]\cdot[[t_4(y,y,y,y)\cdot[y\cdot y]]\cdot x]]]\\
&+ \tfrac{1}{700}\,[y\cdot[t_4(y,y,y,y^2(y^2x))\cdot y]]\\
&+ \tfrac{1}{700}\,[y\cdot[[y\cdot y]\cdot[t_4(y,y,y^2,y^2)\cdot x]]]\\
&+ \tfrac{1}{700}\,[[y\cdot y]\cdot[y\cdot[t_4(y,y,y^2,x)\cdot[y\cdot y]]]]\\
&+ \tfrac{1}{700}\,[t_4(y,y,y(y(yy^2)),x)\cdot[y\cdot y]]\\
&+ \tfrac{1}{700}\,t_4(y,y^2,yy^2,y(y^2x))\\
&+ \tfrac{39}{28000}\,[y\cdot[t_4(y,y,y^2,y^2x)\cdot[y\cdot y]]]\\
&- \tfrac{1}{720}\,[[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot[y\cdot[y\cdot y]]]]\cdot x]\\
&+ \tfrac{1}{720}\,[[t_4(y,y,y,y^2)\cdot[y\cdot[y\cdot[y\cdot y]]]]\cdot x]\\
&+ \tfrac{19}{14000}\,[y\cdot[t_4(y,y,yy^2,x)\cdot[y\cdot[y\cdot y]]]]\\
&+ \tfrac{1}{750}\,[[y\cdot y]\cdot[[t_4(y,y,y,y)\cdot[y\cdot y]]\cdot[y\cdot x]]]\\
&+ \tfrac{1}{750}\,[t_4(y,y,y,y^2)\cdot[[y\cdot y]\cdot[y\cdot[y\cdot x]]]]\\
&+ \tfrac{1}{750}\,[[y\cdot y]\cdot[t_4(y,y,y,y^2x)\cdot[y\cdot y]]]\\
&- \tfrac{1}{750}\,[[y\cdot y]\cdot[t_4(y,y,y^2,y^2)\cdot[y\cdot x]]]
\end{aligned}
$$

$$+ \tfrac{9}{7000}\,[[t_4(y,y,yy^2,x)\cdot y]\cdot[y\cdot[y\cdot y]]]$$
$$- \tfrac{2}{1575}\,[t_4(y,y,y,y)\cdot[[y\cdot y]\cdot[y\cdot[y\cdot[y\cdot x]]]]]$$
$$+ \tfrac{2}{1575}\,[y\cdot[y\cdot[[y\cdot[t_4(y,y,y,y)\cdot y]]\cdot[y\cdot x]]]]$$
$$+ \tfrac{2}{1575}\,[y\cdot[y\cdot[[t_4(y,y,y,y^2)\cdot y]\cdot[y\cdot x]]]]$$
$$+ \tfrac{13}{10500}\,[t_4(y,y,y,y)\cdot[y\cdot[y\cdot[y\cdot[[y\cdot y]\cdot x]]]]]$$
$$- \tfrac{13}{10500}\,[y\cdot[y\cdot[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,y)\cdot x]]]]]$$
$$- \tfrac{13}{10500}\,[[y\cdot y]\cdot[y\cdot[y\cdot[y\cdot[t_4(y,y,y,x)\cdot y]]]]]$$
$$- \tfrac{13}{10500}\,[[y\cdot y]\cdot[y\cdot[y\cdot[t_4(y,y,y,yx)\cdot y]]]]$$
$$- \tfrac{13}{10500}\,[t_4(y,y,y,y^2(y(yx)))\cdot[y\cdot y]]$$
$$+ \tfrac{13}{10500}\,[[t_4(y,y,y^2,yx)\cdot y]\cdot[y\cdot[y\cdot y]]]$$
$$- \tfrac{31}{25200}\,[y\cdot[[y\cdot[t_4(y,y,y,y)\cdot[y\cdot[y\cdot y]]]]\cdot x]]$$
$$- \tfrac{31}{25200}\,[y\cdot[[y\cdot[t_4(y,y,y,y^2)\cdot[y\cdot y]]]\cdot x]]$$
$$+ \tfrac{1}{840}\,[[y\cdot[y\cdot y]]\cdot[t_4(y,y,y,yy^2)\cdot x]]$$
$$+ \tfrac{1}{840}\,[[y\cdot[y\cdot y]]\cdot[[t_4(y,y,y,x)\cdot y]\cdot[y\cdot y]]]$$
$$+ \tfrac{59}{50400}\,[[t_4(y,y,y,y)\cdot[y\cdot[y\cdot y]]]\cdot[[y\cdot y]\cdot x]]$$
$$+ \tfrac{59}{50400}\,[[t_4(y,y,y,y^2)\cdot[y\cdot y]]\cdot[[y\cdot y]\cdot x]]$$
$$- \tfrac{29}{25200}\,[[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot y]]\cdot[y\cdot[y\cdot x]]]$$
$$- \tfrac{29}{25200}\,[[t_4(y,y,y,yy^2)\cdot y]\cdot[[y\cdot y]\cdot x]]$$
$$+ \tfrac{1}{875}\,[y\cdot[[y\cdot[y\cdot y]]\cdot[y\cdot[t_4(y,y,y,y)\cdot x]]]]$$
$$+ \tfrac{1}{875}\,[[y\cdot[y\cdot y]]\cdot[y\cdot[t_4(y,y,y,x)\cdot[y\cdot y]]]]$$
$$- \tfrac{1}{875}\,t_4(y,y,y,y^2(y^2(y^2x)))$$
$$- \tfrac{71}{63000}\,[[t_4(y,y,y,y)\cdot y]\cdot[[y\cdot y]\cdot[y\cdot[y\cdot x]]]]$$
$$- \tfrac{1}{900}\,[[y\cdot[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y)\cdot y]]]]]\cdot x]$$
$$+ \tfrac{1}{900}\,[[t_4(y,y,y,yy^2)\cdot[y\cdot[y\cdot y]]]\cdot x]$$
$$+ \tfrac{23}{21000}\,[y\cdot[t_4(y,y,y,y(y^2x))\cdot[y\cdot y]]]$$
$$- \tfrac{17}{15750}\,[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,y(yx))\cdot y]]]$$
$$- \tfrac{3}{2800}\,t_4(y,y,y^2,y^2(y(y^2x)))$$
$$+ \tfrac{53}{50400}\,[[y\cdot y]\cdot[[t_4(y,y,y,y)\cdot[y\cdot[y\cdot y]]]\cdot x]]$$
$$+ \tfrac{53}{50400}\,[[y\cdot y]\cdot[[t_4(y,y,y,y^2)\cdot[y\cdot y]]\cdot x]]$$
$$- \tfrac{11}{10500}\,[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,y)\cdot[y\cdot[y\cdot x]]]]]$$
$$- \tfrac{13}{12600}\,[[y\cdot[y\cdot y]]\cdot[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot x]]]$$
$$+ \tfrac{13}{12600}\,[y\cdot[y\cdot[[t_4(y,y,y,yy^2)\cdot y]\cdot x]]]$$
$$+ \tfrac{13}{12600}\,[y\cdot[[t_4(y,y^2,y^2,y^2)\cdot y]\cdot x]]$$
$$+ \tfrac{1}{1000}\,[y\cdot[[t_4(y,y,y,y)\cdot[y\cdot y]]\cdot[[y\cdot y]\cdot x]]]$$
$$- \tfrac{1}{1000}\,[y\cdot[t_4(y,y,y^2,y^2)\cdot[[y\cdot y]\cdot x]]]$$
$$- \tfrac{1}{1000}\,[t_4(y,y,yy^2,y^2x)\cdot[y\cdot y]]$$
$$- \tfrac{31}{31500}\,[[y\cdot[t_4(y,y,y,y)\cdot y]]\cdot[y\cdot[y\cdot[y\cdot x]]]]$$
$$- \tfrac{31}{31500}\,[[t_4(y,y,y,y^2)\cdot y]\cdot[y\cdot[y\cdot[y\cdot x]]]]$$

$$
\begin{aligned}
&- \tfrac{1}{1050}\,[[y\cdot y]\cdot[y\cdot[[t_4(y,y,y,x)\cdot y]\cdot[y\cdot y]]]]\\
&+ \tfrac{1}{1050}\,t_4(y,y,y,y^2((y(yy^2))x))\\
&- \tfrac{13}{14000}\,[t_4(y,y,y^2,y^2x)\cdot[y\cdot[y\cdot y]]]\\
&- \tfrac{13}{14000}\,[t_4(y,y^2,y^2,x)\cdot[y\cdot[y\cdot[y\cdot y]]]]\\
&+ \tfrac{23}{25200}\,[t_4(y,y,y,yy^2)\cdot[[y\cdot y]\cdot[y\cdot x]]]\\
&- \tfrac{1}{1120}\,[y\cdot[[y\cdot y]\cdot[y\cdot[[t_4(y,y,y,y)\cdot y]\cdot x]]]]\\
&+ \tfrac{1}{1120}\,[y\cdot[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,y^2)\cdot x]]]]\\
&- \tfrac{37}{42000}\,[t_4(y,y,y,y^2)\cdot[[y\cdot y]\cdot[[y\cdot y]\cdot x]]]\\
&+ \tfrac{11}{12600}\,[[y\cdot[t_4(y,y,y,y)\cdot[y\cdot y]]]\cdot[y\cdot[y\cdot x]]]\\
&- \tfrac{11}{12600}\,[t_4(y,y^2,y^2,y^2)\cdot[y\cdot[y\cdot x]]]\\
&- \tfrac{3}{3500}\,[[y\cdot[y\cdot y]]\cdot[y\cdot[y\cdot[t_4(y,y,y,x)\cdot y]]]]\\
&+ \tfrac{1}{1200}\,[y\cdot[[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot y]]\cdot[y\cdot x]]]\\
&- \tfrac{1}{1200}\,[[y\cdot[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,y)\cdot y]]]]\cdot x]\\
&+ \tfrac{1}{1200}\,[y\cdot[[y\cdot y]\cdot[t_4(y,y,y,yy^2)\cdot x]]]\\
&+ \tfrac{41}{50400}\,[[y\cdot y]\cdot[[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot y]]\cdot x]]\\
&- \tfrac{19}{25200}\,[t_4(y,y,y,y)\cdot[y\cdot[y\cdot[[y\cdot y]\cdot[y\cdot x]]]]]\\
&- \tfrac{19}{25200}\,[[y\cdot[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot y]]]\cdot[y\cdot x]]\\
&- \tfrac{19}{25200}\,[t_4(y,y,y,y^2)\cdot[y\cdot[[y\cdot y]\cdot[y\cdot x]]]]\\
&+ \tfrac{1}{1400}\,[y\cdot[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot[[y\cdot y]\cdot x]]]]\\
&- \tfrac{1}{1400}\,[t_4(y,y,y^2,y(yy^2))\cdot[y\cdot x]]\\
&+ \tfrac{1}{1400}\,[y\cdot[t_4(y,y,y^2,y(yy^2))\cdot x]]\\
&- \tfrac{1}{1400}\,[t_4(y,y,yy^2,yy^2)\cdot[y\cdot x]]\\
&+ \tfrac{1}{1400}\,[y\cdot[t_4(y,y,yy^2,yy^2)\cdot x]]\\
&+ \tfrac{1}{1400}\,[t_4(y,y,yy^2,y(y^2x))\cdot y]\\
&+ \tfrac{1}{1400}\,[t_4(y,y,y(yy^2),x)\cdot[y\cdot[y\cdot y]]]\\
&- \tfrac{29}{42000}\,[y\cdot[y\cdot[[t_4(y,y,y,y)\cdot y]\cdot[[y\cdot y]\cdot x]]]]\\
&- \tfrac{43}{63000}\,[[y\cdot y]\cdot[t_4(y,y,y,y^2)\cdot[y\cdot[y\cdot x]]]]\\
&+ \tfrac{17}{25200}\,[[y\cdot[t_4(y,y,y,y)\cdot[y\cdot[y\cdot y]]]]\cdot[y\cdot x]]\\
&+ \tfrac{17}{25200}\,[[y\cdot[t_4(y,y,y,y^2)\cdot[y\cdot y]]]\cdot[y\cdot x]]\\
&+ \tfrac{1}{1500}\,[[y\cdot y]\cdot[t_4(y,y,y,y^2)\cdot[[y\cdot y]\cdot x]]]\\
&+ \tfrac{1}{1500}\,[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,y^2x)\cdot y]]]\\
&- \tfrac{1}{1575}\,[[y\cdot y]\cdot[y\cdot[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot x]]]]\\
&- \tfrac{1}{1575}\,[[y\cdot[t_4(y,y,y,y)\cdot y]]\cdot[[y\cdot y]\cdot[y\cdot x]]]\\
&- \tfrac{1}{1575}\,[[t_4(y,y,y,y^2)\cdot y]\cdot[[y\cdot y]\cdot[y\cdot x]]]\\
&+ \tfrac{1}{1575}\,[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,y^2)\cdot[y\cdot x]]]]\\
&+ \tfrac{53}{84000}\,[y\cdot[y\cdot[[y\cdot y]\cdot[[t_4(y,y,y,y)\cdot y]\cdot x]]]]\\
&- \tfrac{13}{21000}\,[[y\cdot y]\cdot[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y)\cdot x]]]]]\\
&- \tfrac{13}{21000}\,t_4(y,y^2,y^2,y^2(y(yx)))
\end{aligned}
$$

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\begin{aligned}
&- \tfrac{19}{31500}\,[y\cdot[[y\cdot[y\cdot[t_4(y,y,y,y)\cdot[y\cdot y]]]]\cdot x]]\\
&- \tfrac{1}{1680}\,[t_4(y,y,y,y)\cdot[[y\cdot[y\cdot[y\cdot[y\cdot y]]]]\cdot x]]\\
&+ \tfrac{1}{1680}\,[y\cdot[[t_4(y,y,y^2,yy^2)\cdot y]\cdot x]]\\
&+ \tfrac{37}{63000}\,[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot[y\cdot[y\cdot[y\cdot x]]]]]\\
&- \tfrac{37}{63000}\,[y\cdot[[y\cdot[t_4(y,y,y,yy^2)\cdot y]]\cdot x]]\\
&- \tfrac{73}{126000}\,[y\cdot[[y\cdot[y\cdot y]]\cdot[t_4(y,y,y,y)\cdot[y\cdot x]]]]\\
&+ \tfrac{29}{50400}\,[[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot y]]\cdot[[y\cdot y]\cdot x]]\\
&+ \tfrac{1}{1750}\,[[y\cdot y]\cdot[y\cdot[[t_4(y,y,y,y)\cdot[y\cdot y]]\cdot x]]]\\
&+ \tfrac{1}{1750}\,[[y\cdot[y\cdot y]]\cdot[y\cdot[t_4(y,y,y,y^2)\cdot x]]]\\
&- \tfrac{1}{1750}\,[t_4(y,y,y,y(y(y^2x)))\cdot[y\cdot y]]\\
&- \tfrac{1}{1750}\,[[y\cdot y]\cdot[y\cdot[t_4(y,y,y^2,y^2)\cdot x]]]\\
&+ \tfrac{1}{1750}\,[t_4(y,y,y^2,y(y^2x))\cdot[y\cdot y]]\\
&+ \tfrac{1}{1750}\,[t_4(y,y,yy^2,x)\cdot[y\cdot[y\cdot[y\cdot y]]]]\\
&- \tfrac{1}{1750}\,[[t_4(y,y^2,y^2,x)\cdot y]\cdot[y\cdot[y\cdot y]]]\\
&+ \tfrac{1}{1800}\,[[[y\cdot[t_4(y,y,y,y)\cdot y]]\cdot[y\cdot[y\cdot y]]]\cdot x]\\
&+ \tfrac{1}{1800}\,[[[t_4(y,y,y,y^2)\cdot y]\cdot[y\cdot[y\cdot y]]]\cdot x]\\
&- \tfrac{13}{25200}\,[y\cdot[y\cdot[[[t_4(y,y,y,y)\cdot y]\cdot[y\cdot y]]\cdot x]]]\\
&- \tfrac{4}{7875}\,[[y\cdot[y\cdot[t_4(y,y,y,y)\cdot[y\cdot y]]]]\cdot[y\cdot x]]\\
&+ \tfrac{1}{2000}\,[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,y)\cdot[[y\cdot y]\cdot x]]]]\\
&- \tfrac{1}{2000}\,[[y\cdot[y\cdot[y\cdot y]]]\cdot[t_4(y,y,y,y)\cdot[y\cdot x]]]\\
&- \tfrac{1}{2000}\,[[y\cdot[y\cdot[y\cdot y]]]\cdot[y\cdot[t_4(y,y,y,y)\cdot x]]]\\
&- \tfrac{1}{2100}\,[y\cdot[[y\cdot y]\cdot[[y\cdot[t_4(y,y,y,y)\cdot y]]\cdot x]]]\\
&- \tfrac{1}{2100}\,[[y\cdot y]\cdot[y\cdot[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot x]]]]\\
&- \tfrac{1}{2100}\,[y\cdot[[y\cdot y]\cdot[[t_4(y,y,y,y^2)\cdot y]\cdot x]]]\\
&- \tfrac{1}{2100}\,[[t_4(y,y,y,yx)\cdot[y\cdot y]]\cdot[y\cdot[y\cdot y]]]\\
&+ \tfrac{1}{2100}\,t_4(y,y,y,y(y(y^2(y^2x))))\\
&- \tfrac{13}{28000}\,[y\cdot[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot[[y\cdot y]\cdot x]]]]\\
&+ \tfrac{1}{2250}\,[[y\cdot y]\cdot[[y\cdot[t_4(y,y,y,y)\cdot y]]\cdot[y\cdot x]]]\\
&+ \tfrac{1}{2250}\,[[y\cdot y]\cdot[[t_4(y,y,y,y^2)\cdot y]\cdot[y\cdot x]]]\\
&+ \tfrac{3}{7000}\,[[t_4(y,y,y,y)\cdot[y\cdot y]]\cdot[y\cdot[[y\cdot y]\cdot x]]]\\
&- \tfrac{3}{7000}\,[t_4(y,y,y,y^2)\cdot[y\cdot[y\cdot[[y\cdot y]\cdot x]]]]\\
&+ \tfrac{3}{7000}\,[[t_4(y,y,y,x)\cdot[y\cdot y]]\cdot[y\cdot[y\cdot[y\cdot y]]]]\\
&+ \tfrac{3}{7000}\,[t_4(y,y,y,y^2x)\cdot[y\cdot[y\cdot[y\cdot y]]]]\\
&- \tfrac{3}{7000}\,[t_4(y,y,y,y^2(y^2x))\cdot[y\cdot y]]\\
&- \tfrac{3}{7000}\,[t_4(y,y,y^2,y^2)\cdot[y\cdot[[y\cdot y]\cdot x]]]\\
&- \tfrac{3}{7000}\,[y\cdot[[y\cdot[t_4(y,y,y^2,y^2)\cdot y]]\cdot x]]\\
&+ \tfrac{3}{7000}\,[[y\cdot[t_4(y,y,y^2,y^2)\cdot y]]\cdot[y\cdot x]]\\
&+ \tfrac{3}{7000}\,t_4(y,y,yy^2,y^2(y^2x))
\end{aligned}
$$

$$+ \tfrac{3}{7000}\, t_4(y, y^2, y^2, y(y(y^2 x)))$$
$$- \tfrac{1}{2625}\, [[y \cdot [y \cdot y]] \cdot [y \cdot [[t_4(y, y, y, y) \cdot y] \cdot x]]]$$
$$+ \tfrac{1}{2625}\, [y \cdot [[y \cdot [y \cdot [t_4(y, y, y, y^2) \cdot y]]] \cdot x]]$$
$$- \tfrac{1}{2625}\, [[y \cdot [y \cdot [t_4(y, y, y, y^2) \cdot y]]] \cdot [y \cdot x]]$$
$$+ \tfrac{1}{2625}\, [[y \cdot y] \cdot [y \cdot [t_4(y, y, y, y y^2) \cdot x]]]$$
$$- \tfrac{23}{63000}\, [[y \cdot y] \cdot [[t_4(y, y, y, y) \cdot y] \cdot [y \cdot [y \cdot x]]]]$$
$$+ \tfrac{1}{2800}\, [t_4(y, y, y, y) \cdot [[y \cdot y] \cdot [y \cdot [[y \cdot y] \cdot x]]]]$$
$$+ \tfrac{1}{3000}\, [y \cdot [[y \cdot [t_4(y, y, y, y) \cdot y]] \cdot [[y \cdot y] \cdot x]]]$$
$$+ \tfrac{1}{3000}\, [y \cdot [[t_4(y, y, y, y^2) \cdot y] \cdot [[y \cdot y] \cdot x]]]$$
$$+ \tfrac{1}{3000}\, t_4(y, y, y, (y(y y^2))(y^2 x))$$
$$+ \tfrac{1}{3150}\, [[y \cdot y] \cdot [y \cdot [t_4(y, y, y, y x) \cdot [y \cdot y]]]]$$
$$+ \tfrac{13}{42000}\, [[y \cdot y] \cdot [t_4(y, y, y, y) \cdot [y \cdot [[y \cdot y] \cdot x]]]]$$
$$+ \tfrac{13}{42000}\, [[y \cdot [y \cdot y]] \cdot [t_4(y, y, y, y) \cdot [[y \cdot y] \cdot x]]]$$
$$+ \tfrac{1}{3500}\, [[t_4(y, y, y, y) \cdot y] \cdot [y \cdot [y \cdot [[y \cdot y] \cdot x]]]]$$
$$- \tfrac{1}{3500}\, [y \cdot [[t_4(y, y, y, y y^2) \cdot [y \cdot y]] \cdot x]]$$
$$+ \tfrac{1}{3500}\, [[t_4(y, y, y, y y^2) \cdot [y \cdot y]] \cdot [y \cdot x]]$$
$$+ \tfrac{1}{3500}\, t_4(y, y, y y^2, y(y(y^2 x)))$$
$$- \tfrac{1}{4200}\, [y \cdot [[t_4(y, y, y, y) \cdot [y \cdot [y \cdot [y \cdot y]]]] \cdot x]]$$
$$+ \tfrac{1}{4200}\, [y \cdot [[[t_4(y, y, y, y) \cdot y] \cdot [y \cdot [y \cdot y]]] \cdot x]]$$
$$+ \tfrac{1}{4200}\, [[t_4(y, y, y, y) \cdot [y \cdot [y \cdot [y \cdot y]]]] \cdot [y \cdot x]]$$
$$- \tfrac{1}{4200}\, [[[t_4(y, y, y, y) \cdot y] \cdot [y \cdot [y \cdot y]]] \cdot [y \cdot x]]$$
$$- \tfrac{1}{4200}\, [y \cdot [[t_4(y, y, y, y^2) \cdot [y \cdot [y \cdot y]]] \cdot x]]$$
$$- \tfrac{1}{4200}\, [[y \cdot [y \cdot y]] \cdot [t_4(y, y, y, y^2) \cdot [y \cdot x]]]$$
$$+ \tfrac{1}{4200}\, [[t_4(y, y, y, y^2) \cdot [y \cdot [y \cdot y]]] \cdot [y \cdot x]]$$
$$- \tfrac{1}{4200}\, t_4(y, y, y, (y^2(y(y y^2)))x)$$
$$+ \tfrac{19}{84000}\, [y \cdot [[y \cdot [y \cdot y]] \cdot [[t_4(y, y, y, y) \cdot y] \cdot x]]]$$
$$- \tfrac{1}{4500}\, [t_4(y, y, y, y^2) \cdot [y \cdot [y \cdot [y \cdot [y \cdot x]]]]]$$
$$+ \tfrac{3}{14000}\, [[y \cdot [y \cdot [y \cdot y]]] \cdot [t_4(y, y, y, y^2) \cdot x]]$$
$$+ \tfrac{1}{5040}\, [y \cdot [[y \cdot [[t_4(y, y, y, y) \cdot y] \cdot [y \cdot y]]] \cdot x]]$$
$$+ \tfrac{1}{5250}\, [[y \cdot y] \cdot [y \cdot [[y \cdot [t_4(y, y, y, y) \cdot y]] \cdot x]]]$$
$$+ \tfrac{1}{5250}\, [[y \cdot [y \cdot y]] \cdot [y \cdot [y \cdot [t_4(y, y, y, y) \cdot x]]]]$$
$$+ \tfrac{1}{5250}\, [[y \cdot y] \cdot [y \cdot [[t_4(y, y, y, y^2) \cdot y] \cdot x]]]$$
$$- \tfrac{1}{5250}\, [t_4(y, y, y, y y^2) \cdot [y \cdot [[y \cdot y] \cdot x]]]$$
$$- \tfrac{1}{5250}\, [[t_4(y, y, y, y(y^2 x)) \cdot y] \cdot [y \cdot y]]$$
$$+ \tfrac{1}{5600}\, [t_4(y, y, y^2, y^2(y x)) \cdot [y \cdot y]]$$
$$- \tfrac{1}{6000}\, [[y \cdot y] \cdot [[t_4(y, y, y, y) \cdot y] \cdot [[y \cdot y] \cdot x]]]$$
$$- \tfrac{1}{7000}\, [y \cdot [[[y \cdot y] \cdot [y \cdot [t_4(y, y, y, y) \cdot y]]] \cdot x]]$$
$$+ \tfrac{1}{7000}\, [[y \cdot [t_4(y, y, y, y) \cdot y]] \cdot [y \cdot [[y \cdot y] \cdot x]]]$$

$$
\begin{aligned}
&+ \tfrac{1}{7000}\,[[[y\cdot y]\cdot[y\cdot[t_4(y,y,y,y)\cdot y]]]\cdot[y\cdot x]]\\
&+ \tfrac{1}{7000}\,[[t_4(y,y,y,y^2)\cdot y]\cdot[y\cdot[[y\cdot y]\cdot x]]]\\
&+ \tfrac{1}{7000}\,[y\cdot[[t_4(y,y,y,y^2x)\cdot y]\cdot[y\cdot y]]]\\
&- \tfrac{1}{7000}\,t_4(y,y,y,y(y^2(y(y^2x))))\\
&- \tfrac{1}{10500}\,[y\cdot[[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y)\cdot y]]]]\cdot x]]\\
&+ \tfrac{1}{10500}\,[[y\cdot[y\cdot[y\cdot y]]]\cdot[[t_4(y,y,y,y)\cdot y]\cdot x]]\\
&+ \tfrac{1}{10500}\,[[y\cdot[y\cdot[y\cdot[t_4(y,y,y,y)\cdot y]]]]\cdot[y\cdot x]]\\
&+ \tfrac{1}{10500}\,[y\cdot[[[t_4(y,y,y,y^2)\cdot y]\cdot[y\cdot y]]\cdot x]]\\
&- \tfrac{1}{10500}\,[[[t_4(y,y,y,y^2)\cdot y]\cdot[y\cdot y]]\cdot[y\cdot x]]\\
&+ \tfrac{1}{10500}\,[y\cdot[[t_4(y,y,y,y(yx))\cdot y]\cdot[y\cdot y]]]\\
&+ \tfrac{1}{10500}\,t_4(y,y,y,y^2(y(y(y^2x))))\\
&+ \tfrac{1}{12600}\,[[t_4(y,y^2,y^2,y^2)\cdot y]\cdot[y\cdot x]]\\
&- \tfrac{1}{14000}\,[y\cdot[[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot[y\cdot y]]]\cdot x]]\\
&+ \tfrac{1}{14000}\,[[t_4(y,y,y,y)\cdot y]\cdot[[y\cdot y]\cdot[[y\cdot y]\cdot x]]]\\
&+ \tfrac{1}{14000}\,[[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot[y\cdot y]]]\cdot[y\cdot x]]\\
&+ \tfrac{1}{14000}\,[y\cdot[[t_4(y,y,y^2,y^2)\cdot[y\cdot y]]\cdot x]]\\
&- \tfrac{1}{14000}\,[[t_4(y,y,y^2,y^2)\cdot[y\cdot y]]\cdot[y\cdot x]]\\
&+ \tfrac{1}{15750}\,[[y\cdot[t_4(y,y,y,yx)\cdot y]]\cdot[y\cdot[y\cdot y]]]\\
&- \tfrac{1}{16800}\,[[y\cdot y]\cdot[t_4(y,y,y,y)\cdot[[y\cdot y]\cdot[y\cdot x]]]]\\
&- \tfrac{1}{21000}\,[t_4(y,y,y,y)\cdot[y\cdot[[y\cdot y]\cdot[[y\cdot y]\cdot x]]]]\\
&- \tfrac{1}{21000}\,[y\cdot[t_4(y,y,y,yy^2)\cdot[[y\cdot y]\cdot x]]]
\end{aligned}
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